LEARN. ORGANISE. STAY PREPARED.

Chhayapoth

A Bangla-First Agentic Super App for Learning, Everyday Wellbeing and Community Resilience

Proposed team: Team GramNova | **Platforms:** Android & iOS

Abstract

Chhayapoth is a proposed mobile super app that connects Bengali-language assistance with practical daily actions. It brings learning support, health information, income and expense records, weather preparedness and nearby communication into a shared personal workspace. Its agent can help organise study plans, record transactions and configure reminders through permission-based tools. Users can personalise the workspace with widgets and small applications running in a controlled environment. Free-tier-aware model routing and optional authorised provider profiles aim to improve affordability, while local storage keeps selected utilities usable without connectivity. The initial focus is an education-led workflow; wider integration and field evaluation remain development priorities.

1. Problem and intended users

Students and households may need several disconnected tools to understand information, plan activities and maintain records. Language barriers, unfamiliar interfaces and unreliable connectivity can make these tasks harder. Chhayapoth targets rural students, families and local volunteers with a simpler Bengali interface. These needs are design hypotheses to be validated through interviews and a supervised pilot—not completed survey findings.

2. Core modules

Learning	Step-by-step explanations, mathematical graphs, practice questions, feedback and revision plans. Teacher-reviewed material and deterministic checks support accuracy.
Care routines	Reviewed health education, general maternal-health information and user-confirmed reminders based on clinician instructions. No autonomous diagnosis or prescription.
Personal ledger	Manual or conversational income/expense entry, editable categories, accurate totals and export. Ambiguous amounts or dates require clarification.
Preparedness	Source-labelled weather and flood information, timestamps and practical checklists. Expired or cached notices must never appear to be live warnings.
Nearby messaging	Authenticated short-range text communication on compatible devices without internet. Delivery acknowledgements and reconnect handling are required.

Primary workflow: explain a concept → practise → review mistakes → schedule revision → show progress on Home. This offers a focused first release rather than claiming equal maturity across every module.

3. What makes the project different

The agent is intended to do more than answer questions: it connects a request to validated actions in the calendar, ledger and other services. Users can request a study widget or a small specialised app without giving generated code unrestricted control of the phone. The main navigation is **Home, Agent, Calendar and More**, with specialised tools available from the shared workspace.

4. Architecture and responsible automation

Interface → agent orchestrator → permission and validation layer → domain services → local storage/native tools. The same service rules apply to manual and agent-driven actions. Mini-apps require isolation, scoped permissions and network limits. Sensitive records should not be shared across modules or cloud providers by default.

Agent workflows can run in the mobile application while model inference uses a remote provider; this is not a claim of fully offline AI. Routing must respect provider terms, task quality, privacy choices and quotas. Free access is availability-dependent, and paid fallback needs explicit consent. Advanced automation does not remove medical safeguards or operating-system permissions.

5. Safety and reliability

- Health content requires qualified review; reminders do not replace professional care.
- Weather summaries retain source, validity and severity. New warnings need a working update channel.
- Nearby messaging has range, battery and platform limits; it is not a guaranteed rescue network.
- Protect credentials, minimise child and health data, and support correction, deletion and tested backup.

6. Development and evaluation

- Integrate the core data layer, planner and habit tracker with safe migrations.
- Deliver the education-to-calendar workflow before expanding the feature set.
- Test calculations, content, permissions, offline behaviour and real device pairs.
- Propose a four-week, consent-based pilot with 20–30 participants; measure task completion, correctness and usability.

Indicative delivery: discovery → core integration → learning workflow → supporting modules → device testing → supervised pilot.
Illustrative pilot budget: BDT 41,000–85,000 for devices/accessories, connectivity, logistics, review and contingency. This is a planning allowance, not a quotation; salaries and major publication costs are excluded.

7. Expected contribution and current position

The primary alignment is **SDG 4: Quality Education**, supported by SDG 3 through health literacy and SDG 13 through preparedness. These are intended contributions, not demonstrated outcomes. Earlier work explored a native-agent foundation and an interactive prototype; integrated domain features and field effectiveness still require verification. The Chhayapoth source is to be uploaded by the owner.

Repository: https://github.com/AKIBUZZAMAN-AKIB/Chhayapoth

Selected references: WHO: generative AI in health (2024); Google: Nearby Connections; BMD warnings; FFWC; UN 2030 Agenda. Live-feed permissions and platform support must be confirmed before release.